

SPL1 Project Proposal Form, 2025
Institute of Information Technology (IIT)
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Project Description:

CineConnect
A Hybrid Movie Recommendation Tool

CineConnect is a user interactive application that recommends movies to users based on their individual preferences and rating history. The system uses a hybrid recommendation technique that combines **Content Based Filtering** and **Matrix Factorization** to provide accurate and personalized movie suggestions. The system is designed to be user interactive, allowing users to register, rate movies, view their previous ratings, and request movie recommendations through a structured interface.

The application operates using data files instead of databases. Each user can register locally, rate movies, and receive recommendations based on their individual profiles. Registration automatically creates a user specific file in the local storage, where all ratings are stored. Two master files of which one contains metadata (a unique ID, name, director, cast, genres) of movies, while the other one stores all ratings' tuples for matrix factorization.

Content Based Filtering (CBF):
In this method, the system studies the details of each movie (prefiltered) such as its genres, cast, director etc. and compares them with the user's past ratings and preferences. It then finds how similar a new movie is to what the user already likes, using basic similarity calculations. Movies that are most similar to the user's favorite ones are suggested first.

Matrix Factorization (MF):
The system reads the data from ratings file and decomposes the user-movie rating matrix into two smaller matrices one representing latent user preferences and the other representing latent movie characteristics. The method minimizes prediction error and estimates missing ratings, allowing prediction of unseen movies.

Finally, both models are fused to produce a balanced recommendation list. Top-ranked movies are displayed as the system's recommendations.

In addition, CineConnect conceptually supports community building among users with shared movie preferences.

Languages or Tools to be used:

- Language: C, C++
- IDE: Code::Blocks / Visual Studio Code
- Version Control: Git and Github to store

Supervisor's Name: Dr. B M Mainul Hossain

Signature of the supervisor: _____

Date: _____

Before the Midterm Presentation:

I confirm that the progress is satisfactory and I am forwarding it for midterm presentation.

Signature of the supervisor: _____

Date: _____

Midterm Presentation Feedback: